University of Basel

Compositional Processes as Graph
Semantic Modelling and Terminology in the Anton Webern Gesamtausgabe (AWG)

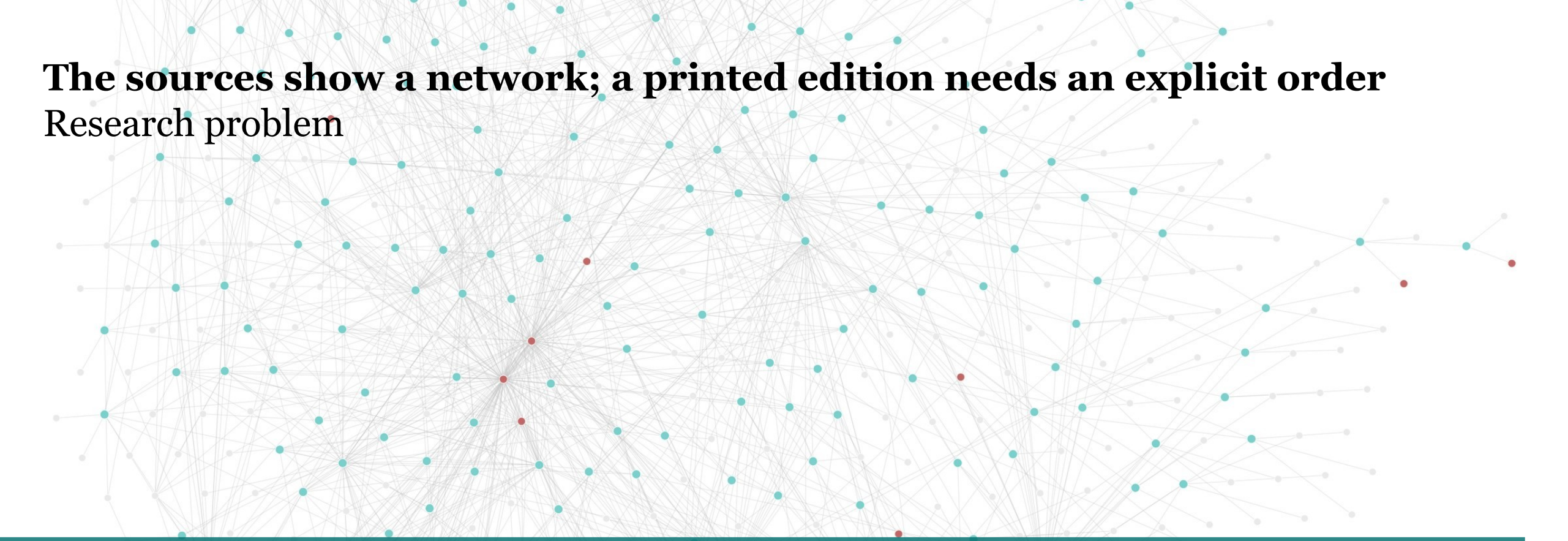


Dr. Stefan Münnich · Anton Webern Gesamtausgabe, University of Basel

“Comparative Perspectives on Genetic Criticism in Music”
International Conference of the IMS Study Group
Rome, German Historical Institute, 16–18 November 2026

The sources show a network; a printed edition needs an explicit order

Research problem

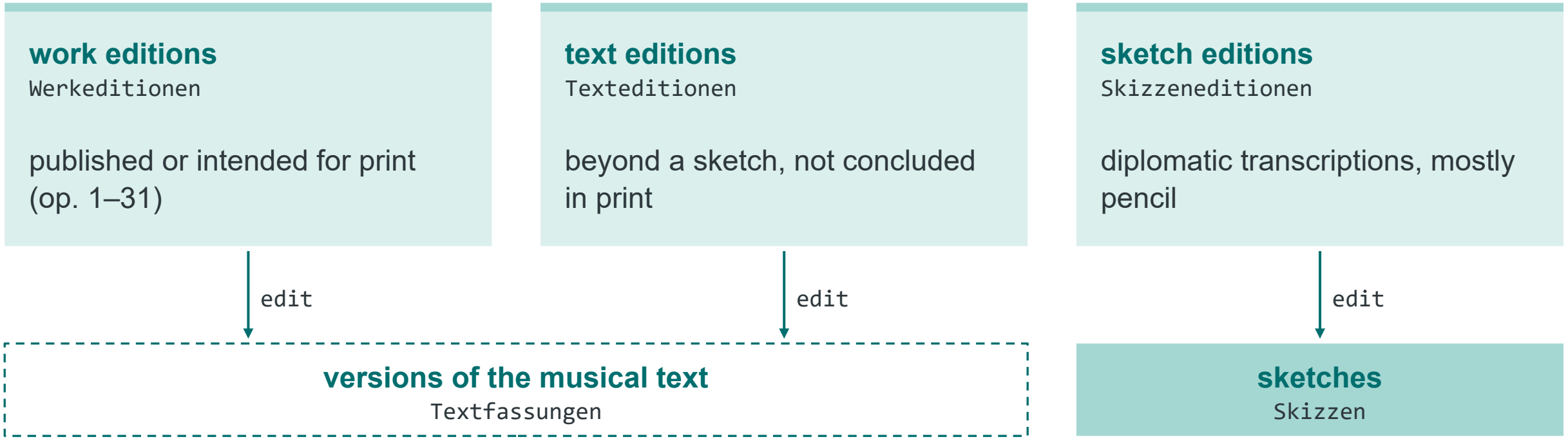


How can a digital edition represent non-linear compositional processes – and keep philological claims as open as the evidence demands?

Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

The AWG distinguishes three forms of edition, by transmission and use

The Anton Webern Gesamtausgabe



a state of the musical text, attested by one or more sources

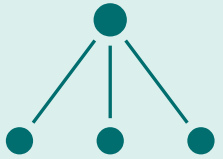
The AWG graph combines generated data with curated editorial assertions

The AWG and its graph

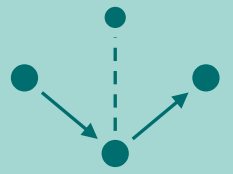
draft ontology v0.1

edition.anton-webern.ch

generated layer
derived from the edition's structured data



curated layer
hand-modelled editorial assertions



Queried with SPARQL in the app, exported as Turtle. No reasoning.

Generated from the edition data alone, op. 25 falls into three “islands”

Graph view: op. 25



1 generated
three “islands”, one per song

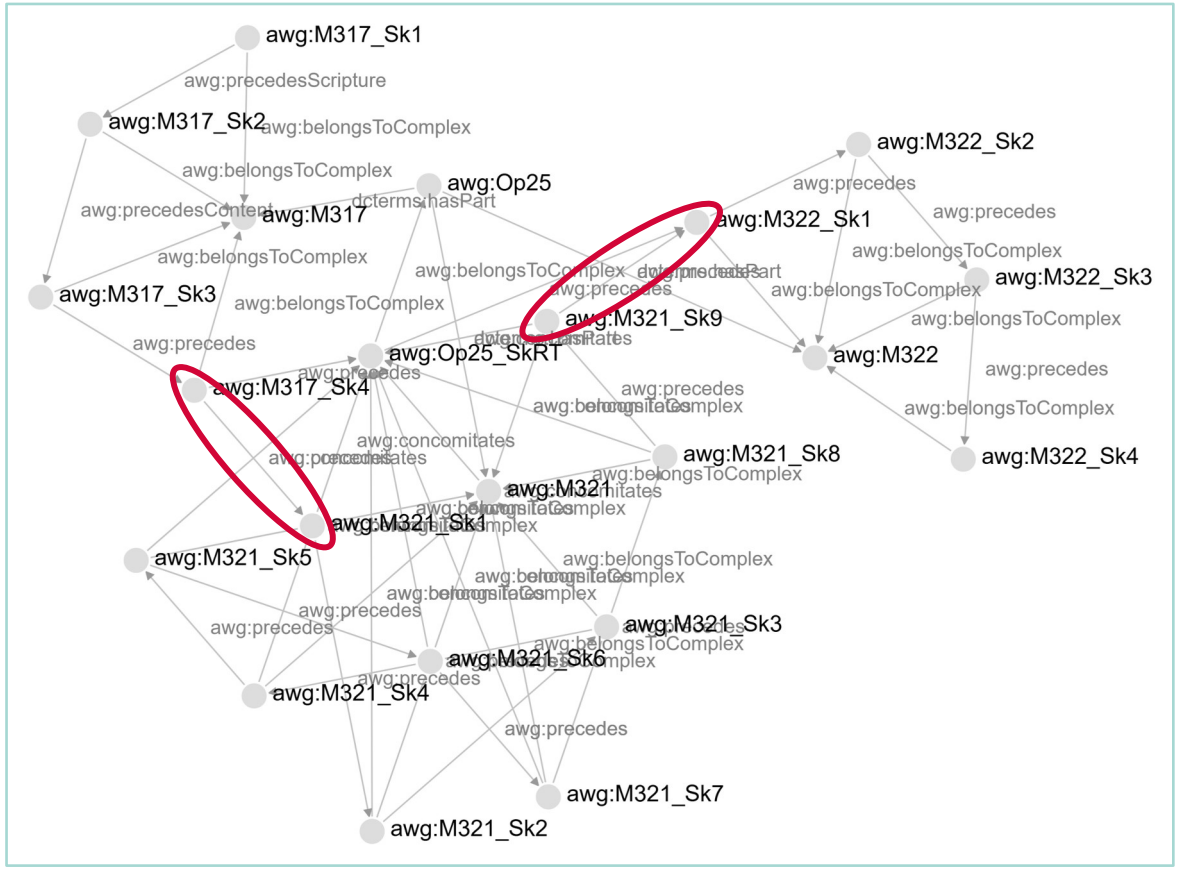
2 curated
one chain across the songs

3 both
the difference: editorial judgment

Source: AWG online edition, op. 25, graph view filtered by SPARQL (development version, screenshots 11 Nov 2026)

The curated layer links the three songs in one chain

Graph view: op. 25



1 generated
three “islands”, one per song

2 curated
one chain across the songs

3 both
the difference: editorial judgment

Source: AWG online edition, op. 25, graph view filtered by SPARQL (development version, screenshots 11 Nov 2026)

The combined layers show where editorial judgment becomes explicit

Graph view: op. 25



1 generated
three “islands”, one per song

2 curated
one chain across the songs

3 both
the difference: editorial judgment

Source: AWG online edition, op. 25, graph view filtered by SPARQL (development version, screenshots 11 Nov 2026)

The categories come from Webern's documents, not from theory

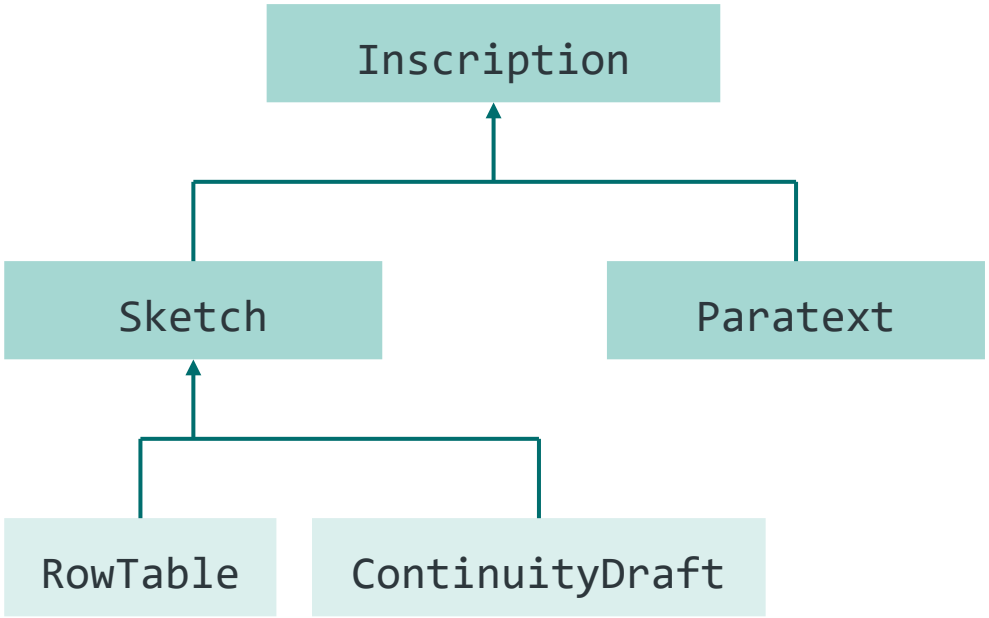
Terms from the material

sketch Skizze	**version** Textfassung	**paratext** Paratext	**row table** Reihentabelle	**edition complex** Editionskomplex
Sk1, Sk2, Sk2.1: the editors' numbering in the source descriptions	TF1, TF2: recognisable states of a song in the sources	title, dating, instrumentation: the categories of the source descriptions	Webern's sheet of row forms	all edition-relevant material of a composition, by opus or M number
in op. 25				
M 317 Sk1–4, M 321 Sk1–9, M 322 Sk1–4, with sub-sketches	M 317 TF2, M 322 TF1, M 321 TF1	title, datings (4. VII. 34), paginations, instrumentations	one sheet, written alongside M 321 Sk1–9	composite: M 317, M 322, M 321 (songs I–III)

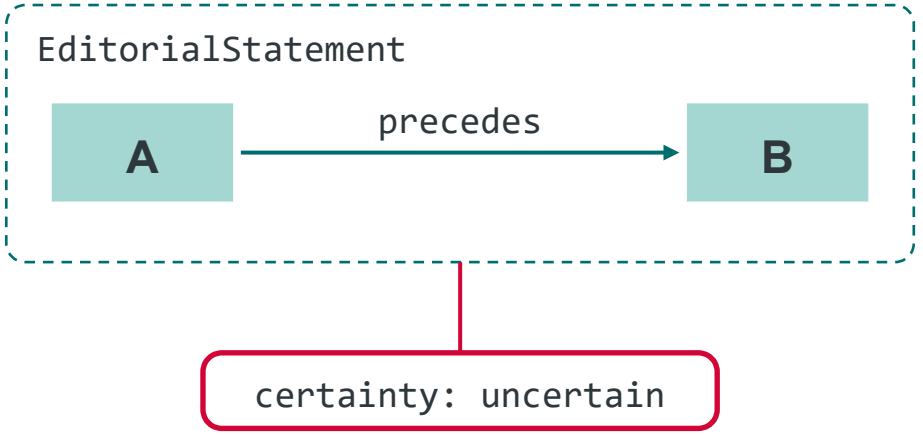
Editorial terms become classes; uncertainty becomes a tag

From term to class

classes



uncertainty

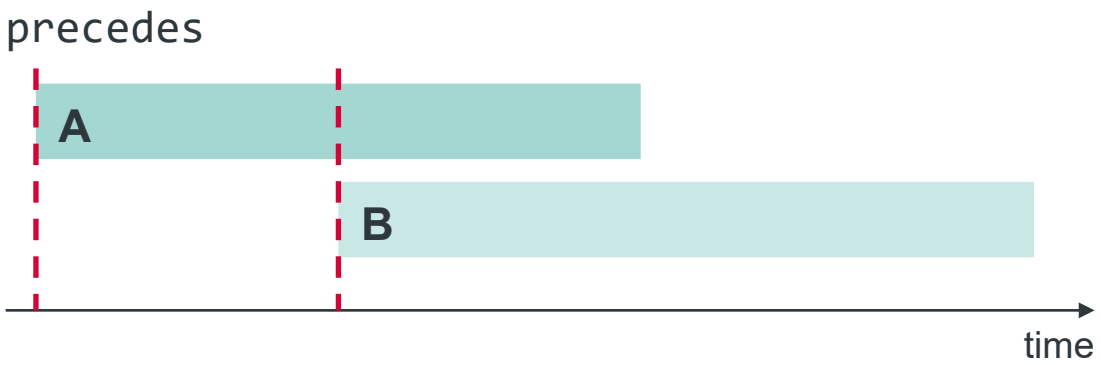


a tag on the statement, not a relation of its own;
without a tag, the statement counts as certain

Classes of the draft ontology v0.1, prefix `awg:` omitted. **Not modelled yet:** “study composition”, “ink draft”.

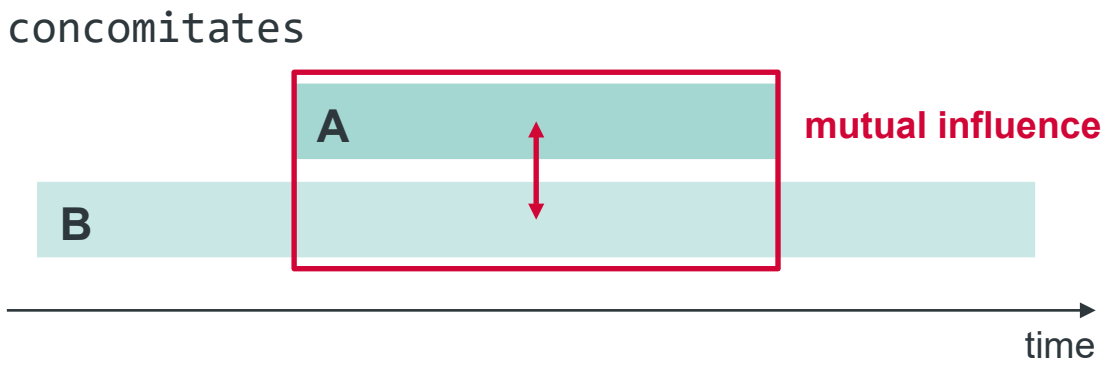
Two relations link inscriptions: a sequence, or mutual influence

Genetic relations



an order can be shown: A begins before B starts (overlap allowed)

cf. CIDOC CRM P176 “starts before the start of”



A and B influence each other; which came first cannot be decided

cf. CIDOC CRM P174 “starts before the end of” (the overlap, not the influence)

evidence: subproperties of `precedes`

`precedesScripture`	`precedesContent`
page, overlay, writing instruments	musical content

For now, the model is domain-specific, with links to standards

Standards and alignments

Relating sketches draws on content and page together – which LRMoo separates (F2 Expression vs. F5 Item).

standard	in the AWG model	link
LRMoo	edition complex ≈ F1 Work · version, sketch ≈ F2 Expression · document ≈ F5 Item	`rdfs:seeAlso`
CIDOC CRM	temporal reading: precedes ≈ P176, concomitates ≈ P174	`rdfs:seeAlso`
Dublin Core terms, SKOS, PROV, `rdf:Statement`	single terms: parts, concept schemes, attribution, statements	`used directly`
MEI	not integrated yet 😢	–

The model could transfer; the pipeline cannot

What transfers

portable	
`Inscription`	what is written and ordered
`Document`	the physical object
`Folio`	one leaf (recto and verso)
`Placement`	where on recto or verso: staves, measures
`precedes`	temporal sequence, overlap allowed
`concomitates`	mutual influence, no order

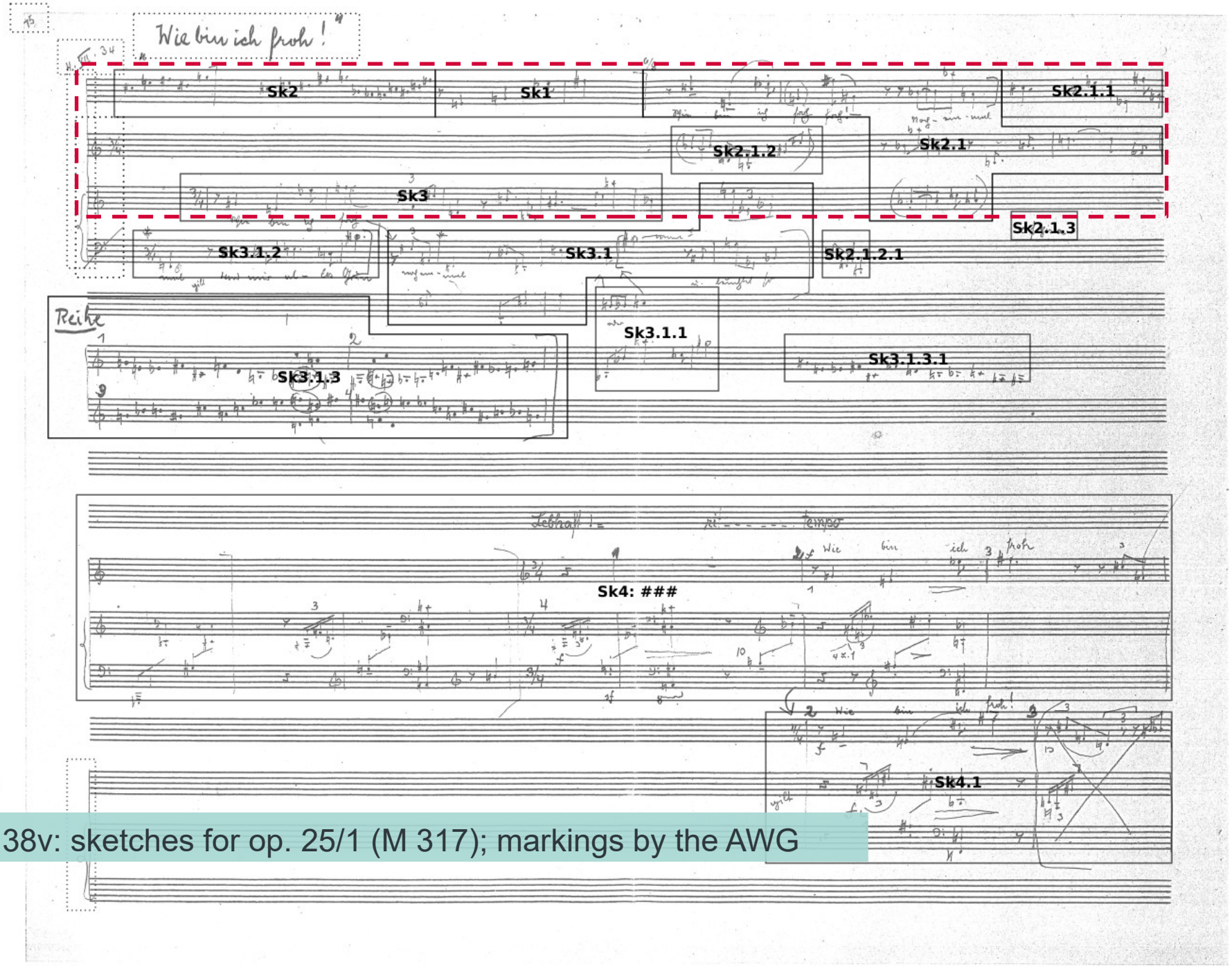
tied to the AWG

the pipeline
the generator reads the AWG's own data

the Moldenhauer catalogue
M numbers name the complexes

the numbering conventions
Sk1 before Sk2 is read as precedes, without evidence type

Transfer to other repertoires is plausible but untested.

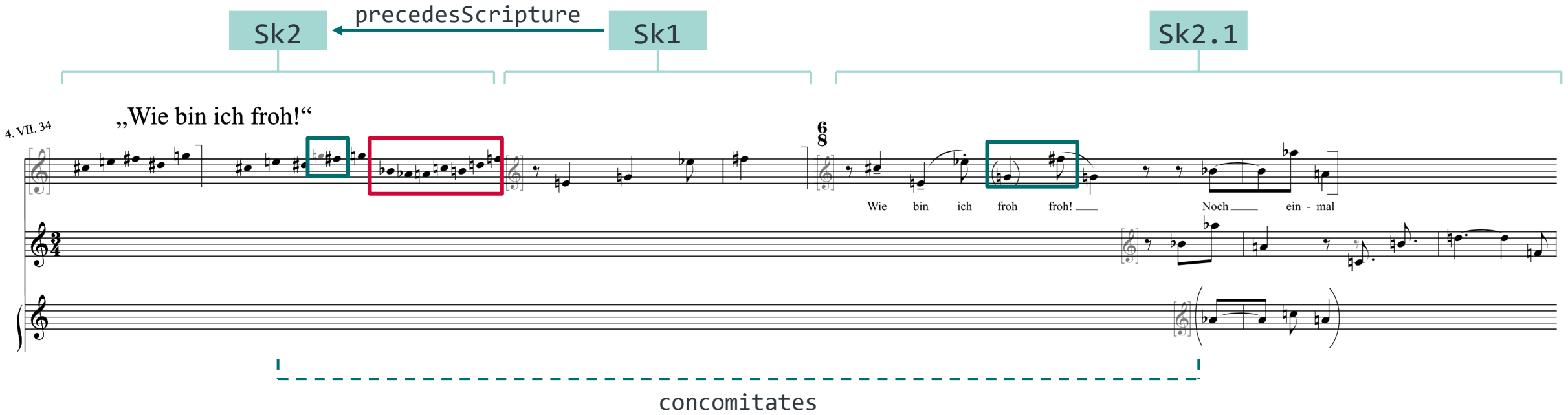


Sketchbook 3, fol. 38v: sketches for op. 25/1 (M 317); markings by the AWG

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On one sketchbook page, both relations are at work

Example: M 317, fol. 38v



Sk1 precedesScripture Sk2

The right end of Sk2 is squeezed together (red): Sk1 already took the space. An order inferred from the writing.

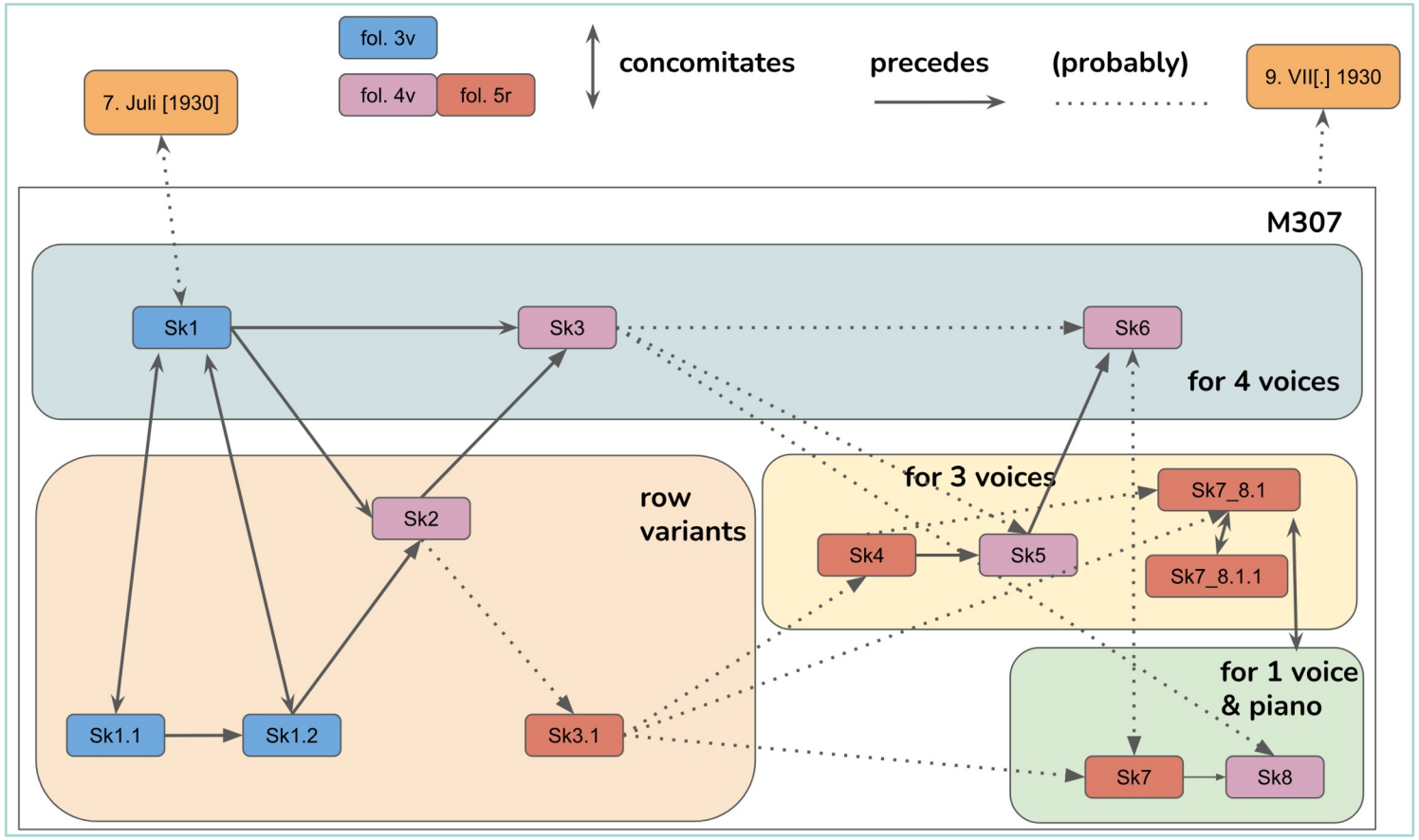
Sk2.1 concomitates Sk2

In both, g gives way to f♯ (green): Sk2.1 and row B of Sk2 shape each other. Which came first cannot be decided.

Source: AWG online edition, M 317 Sk2, Sk1, Sk2.1, source op. 25:A, fol. 38v, systems 1–3, superimposed (development version, 12 Nov 2026)

A process Webern kept reordering, and then dropped, becomes graspable

Exploring the graph: M 307



Source: Stefan Münnich/Thomas Ahrend, own graphic (2019): sketch graph of M 307

the scoring shifts

four voices, three voices, voice and piano – row variants in between

across the pages

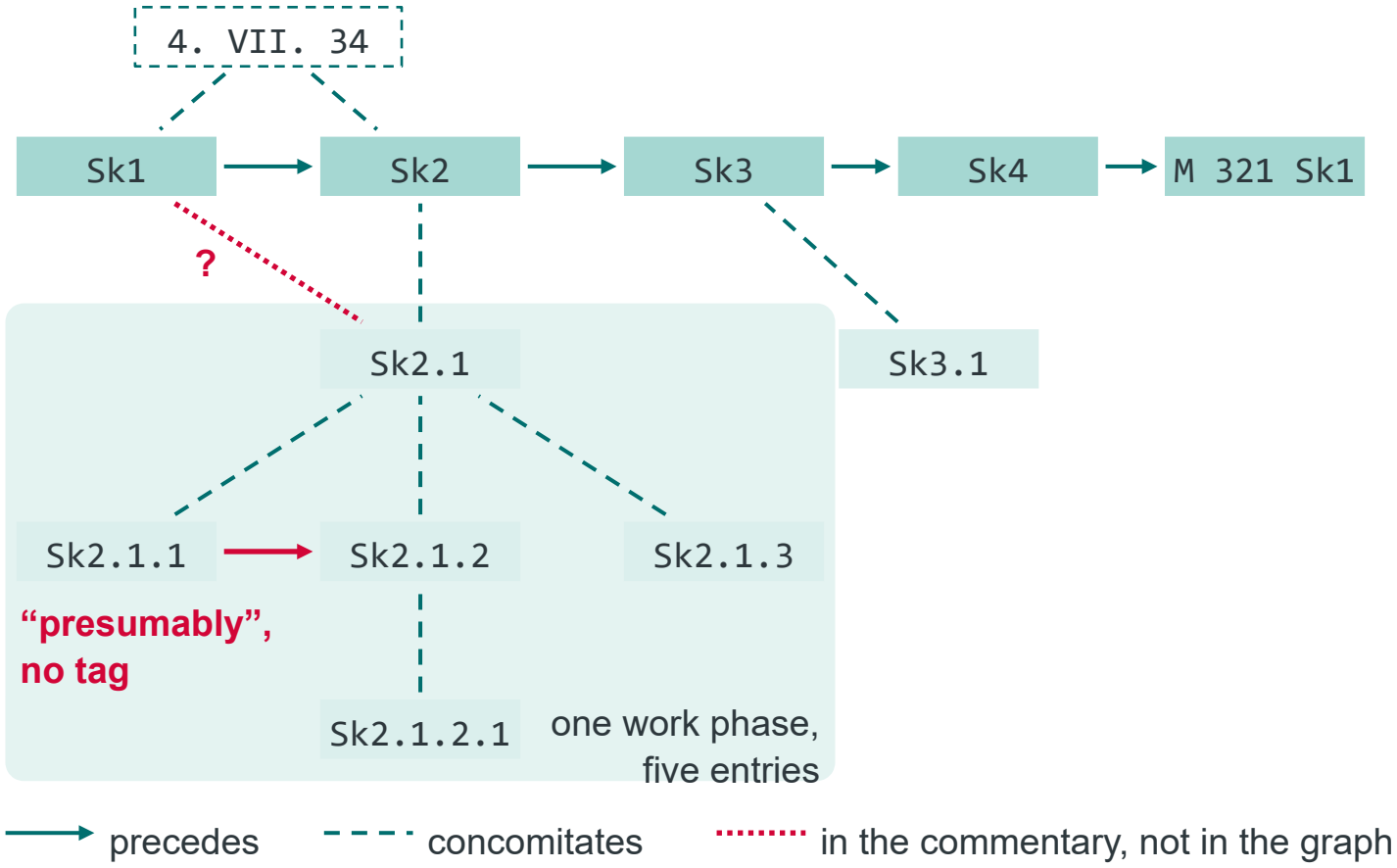
sketches of fol. 3v, 4v and 5r, ordered across the groups of scoring

for the editors

a way into a process that Webern finally abandoned

The graph shows patterns the commentary spreads out – and gaps it hides

Exploring the graph: M 317



→ Clusters

Sk2.1 and its variants: one work phase at a glance.

→ Paths

From Sk1 to Sk4 and on into M 321: the work leaves the song.

→ Gaps

Sk2.1 builds on Sk1, but no edge says so. A “presumably” has no tag.

The graph does not discover. It makes commentary and data comparable.

Selection from the curated layer of the op. 25 graph (M 317), drawn from the data (development version, 12 Nov 2026)

Where model and data diverge, each gap is information

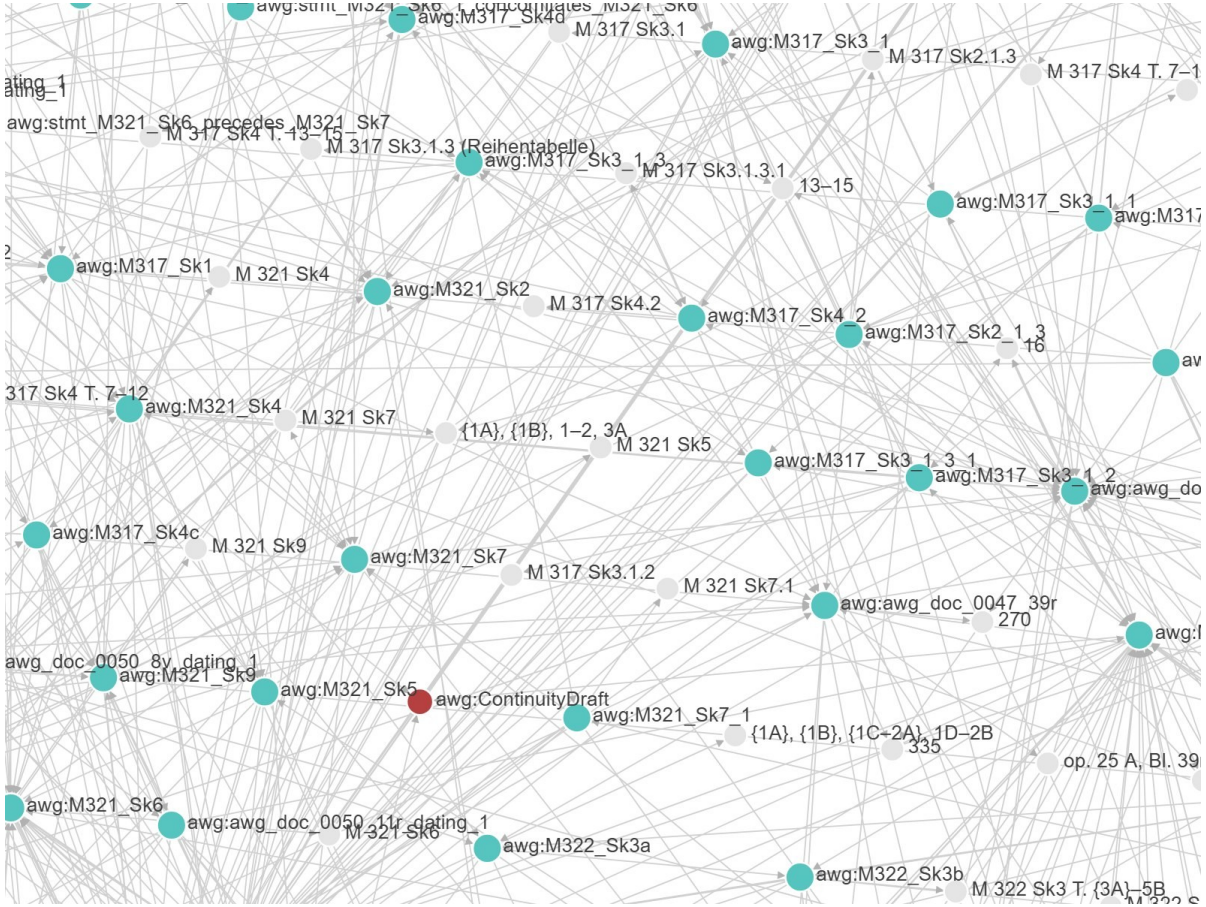
Frictions: model vs. data

model lags data	data lags model
staves own notation: “1c–3c”	**sketchbooks** per complex: sketchbook 3 as two excerpts
measures own notation: “{1A}, {1B}, 1–2, 3A”	Sk## **placeholders** `warning: placeholder` 29 sketches without stable identifier
writing instruments missing: pencil, ink, crayon	**incomplete dates** `warning: incompleteDate` “M. am 30. IX. …”, no year

Some gaps are ours to close; others belong to the evidence.

The mismatch between model and data is the analytical gain

Conclusion and outlook



Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

> Layers show where editorial judgment becomes explicit.

> Warnings show where the data are open or disagree.

> Model and data are written side by side: concomitant.

Next steps: named graphs for ink and pencil layers, persons, archives

If model and data shape each other, the graph is also our construction

Closing question

Whose process does the graph show – Webern's or ours?

How much of (Webern's) compositional processes survives translation into formal models?

Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

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Thank you

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